

Emotion Classification in NFL Tweets Using Pseudo-Labeling and Classical ML

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May 2026

Abstract

This report presents an emotion classification pipeline for short, informal NFL-related tweets. A corpus of approximately 6,200 tweets was scraped and automatically labeled with six emotion categories using a pre-trained DistilRoBERTa model. Four classical classifiers (Logistic Regression, Linear SVC, Complement Naive Bayes, SGD Classifier) were trained on combined word- and character-level TF-IDF features, compared via 5-fold cross-validation, and tuned with GridSearchCV. Error analysis reveals that pseudo-label noise and semantic overlap between emotions are the main bottlenecks. The project code is available at https://github.com/grottoheaven/nlp_proj.

1 Introduction

Emotion detection in text has become a crucial task in natural language processing (NLP), with applications ranging from social media monitoring to customer feedback analysis and brand sentiment tracking. The rise of deep learning and transfer learning has significantly advanced the field, yet there remains a practical need for interpretable and computationally efficient models—especially when dealing with domain-specific data for which large manually annotated corpora do not exist. This work focuses on emotion classification in short, informal texts (tweets) related to American football and the National Football League (NFL), a domain rich in emotional expression, from elation after a last-minute touchdown to fury following a controversial referee call.

The primary goal is to investigate whether classical machine learning classifiers, trained on TF-IDF features extracted from pseudo-labeled data, can serve as effective baselines for emotion recognition in this niche domain. A custom **Selenium** scraper collected approximately 6,200 tweets, which were automatically annotated using the pre-trained DistilRoBERTa model from HuggingFace. After excluding neutral tweets, six emotion categories remain: *anger*, *disgust*, *fear*, *happiness*, *sadness*, and *surprise*. Four classifiers (Logistic Regression, Linear SVC, Complement Naive Bayes, and SGD Classifier) were trained on a combination of word-level unigrams/bigrams and character-level n-grams, and

thoroughly evaluated via cross-validation, grid search, and held-out test set. An in-depth error analysis highlights systematic misclassification patterns caused by pseudo-label noise, semantic overlap, and class imbalance.

1.1 Team

Maxim Afonin is the only member.

2 Related Work

Emotion detection from text has evolved from lexicon-based methods to deep neural architectures. Early works used hand-crafted features and rule-based systems; modern approaches leverage pre-trained language models. [1] survey the shift toward deep learning and transformers. [2] propose LRA-DNN for efficient social media emotion analysis.

For short-text emotion classification, [3] and [4] offer improved multi-label techniques. BERT-based architectures have set new standards: [5] show RoBERTa’s strength for context-based emotion detection. Other notable works include evaluations of large language models for emotion understanding [6].

Despite these advances, most studies use general-domain datasets. The NFL domain introduces unique vocabulary (team names, jargon) that challenges generic models. This work bridges the gap by applying a transformer-based pseudo-labeling strategy combined with lightweight classical classifiers and provides a thorough error analysis.

3 Model Description

The approach consists of three main stages: data collection and pseudo-labeling, feature extraction, and classification.

3.1 Data Collection and Pseudo-Labeling

Tweets were gathered with a headless **Selenium** browser, searching NFL-related keywords (team names, players, events) with filters for English language and minimum engagement. After deduplication, about 6,200 tweets were collected. Manual annotation was impractical, so pseudo-labels were generated using the **j-huggingface/emotion-english-distil-roberta-base** model from Hugging-Face. It predicts seven categories: *anger*, *disgust*, *fear*, *joy*, *neutral*, *sadness*, and *surprise*. The *neutral* class was removed, and *joy* was renamed *happiness*, leaving six emotions.

3.2 TF-IDF Feature Extraction

Texts were lowercased, URLs and mentions removed, and hashtag symbols stripped. A **TfidfVectorizer** extracted two feature sets:

- **Word-level:** unigrams and bigrams, max 5,000 features, min. document frequency 5.
- **Character-level:** char n-grams of length 2–4, max 3,000 features.

The two sparse matrices were concatenated into an $N \times 8000$ feature matrix.

3.3 Classification Models

Four classical classifiers were trained:

- **Logistic Regression** (saga solver, ℓ_2 penalty, balanced class weights).
- **Linear SVC** (balanced class weights).
- **Complement Naive Bayes** ($\alpha = 0.1$).
- **SGD Classifier** (modified Huber loss, balanced weights).

The two best models (Logistic Regression and Linear SVC) were tuned with GridSearchCV over $C \in \{0.01, 0.1, 0.5, 1.0, 5.0, 10.0\}$ and penalty type for Logistic Regression.

4 Dataset

The collected corpus contains approximately 6,200 tweets related to the NFL. After removing duplicates and tweets labeled as *neutral*, the working dataset comprises tweets in six emotion categories. The approximate distribution is: *surprise* $\sim 38\%$, *happiness* $\sim 27\%$, *sadness* $\sim 14\%$, *anger* $\sim 10\%$, *disgust* $\sim 7\%$, *fear* $\sim 4\%$. The dataset is split into training (85%), and test (15%) sets using stratified sampling. Table 1 summarises the split.

Table 1: Dataset split statistics. The class imbalance is clearly visible.

Split	Samples	Majority class (<i>surprise</i>)	Minority class (<i>fear</i>)
Train	2,980	1,133	133
Test	527	200	24

The dataset is not publicly released, but it was collected exclusively from publicly available tweets and processed for research purposes. Pseudo-labels were generated automatically; a small set of human-annotated tweets is planned for future quality assessment.

5 Experiments

5.1 Metrics

Performance is evaluated using accuracy, macro-averaged precision, recall, and F1-score, as well as weighted F1. Macro-averaging treats all classes equally, which is appropriate given the severe class imbalance. Per-class F1 is also reported to highlight minority class performance.

5.2 Experiment Setup

All experiments were run in Python using scikit-learn and HuggingFace transformers. A 5-fold stratified cross-validation was performed on the training set. Hyperparameters for Logistic Regression and Linear SVC were tuned via GridSearchCV with 5-fold CV on the training set, guided by the validation set. The final evaluation was conducted on the held-out test set (527 tweets). All models used balanced class weights where applicable.

5.3 Baselines

The majority-class baseline (always predicting *surprise*) yields an accuracy of $\sim 38\%$ and a macro-F1 of approximately 0.167 (since only one class is predicted). Our classifiers are also compared against each other to identify the most suitable linear model for this pseudo-labeled, sparse representation setting.

6 Results

Table 2 reports the final test set performance. Complement Naive Bayes achieves the highest macro-F1 (0.455), surpassing even the tuned logistic regression (0.442). Linear SVC, the cross-validation leader, reaches 0.426. All models significantly outperform the majority baseline.

Table 2: Test set performance (sorted by macro-F1).

Model	Accuracy	F1-macro	F1-weighted	Precision (macro)	Recall (macro)
Complement Naive Bayes	0.535	0.455	0.533	0.465	0.448
Logistic Regression (tuned)	0.541	0.442	0.537	0.452	0.436
SGD Classifier	0.545	0.427	0.528	0.522	0.403
LinearSVC (tuned)	0.529	0.426	0.526	0.430	0.423

The per-class F1 scores for the best cross-validation model (LinearSVC) are visualized in Fig. 1. The dominant classes (*happiness*, *sadness*, *surprise*) exceed 0.5 F1, while *anger* (0.30) and *fear* (0.19) remain challenging.

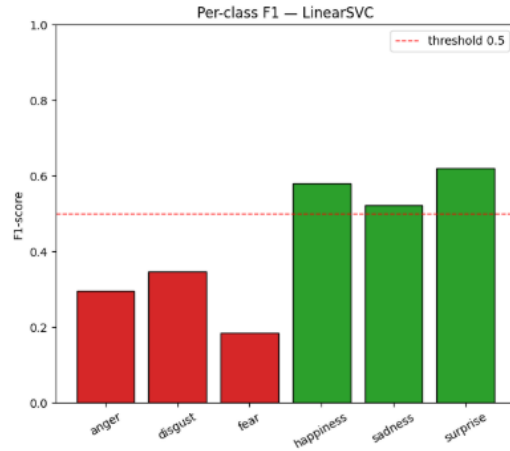


Figure 1: Per-class F1 scores of the tuned LinearSVC on the test set.

The confusion matrix (Fig. 2) reveals that *anger* and *sadness* are often confused with *surprise*, likely due to shared exclamatory vocabulary and the teacher model’s labeling bias.

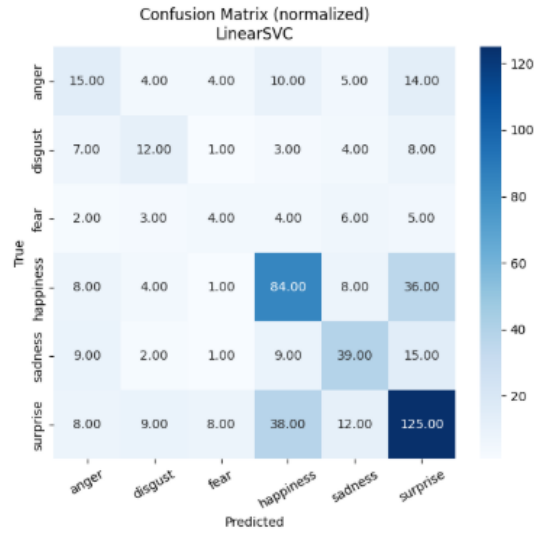


Figure 2: Confusion matrix (raw counts) of the tuned LinearSVC on the test set.

7 Conclusion

This work presented a complete pipeline for domain-specific emotion classification in NFL tweets. By combining automatic pseudo-labeling with a DistilRoBERTa teacher model and training classical linear classifiers on TF-IDF features, we achieved a macro-F1 of up to 0.455, significantly above the majority baseline. Complement Naive Bayes emerged as the best test-set performer, while error analysis highlighted that pseudo-label noise and confusion between adjacent emotions (especially *anger* and *sadness* with *surprise*) are the primary limitations. Future work should incorporate a small human-annotated validation set, explore contextual embeddings as features, and investigate multi-label classification to better capture the complex emotional landscape of sports-focused social media.

References

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